



Adi Menon

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Website Home 1 min pitch

STANDARDIZED TESTS

SAT
R/W 740, Math 720, 2026

AP Scholar, College Board, 2026

EDUCATION

Pacific Collegiate School, Santa Cruz, CA Class of 2027
GPA 4.34 weighted, weighted 3.83,

12th: AP English Literature, AP Physics C (E&M), AP Microeconomics, AP Statistics, Topics in Design & Engineering (in progress)

11th: AP Computer Science A (A/A), AP English Language (A/A), Multivariable Calculus Honors (A/B), AP Physics C Mechanics (A/A), AP United States History (A/A)

10th: AP Calculus BC (A/A), Chemistry (A/A), English 10 (A/A), AP Spanish (A/B), AP World History (A/A), Video Production 2 (A/A)

9th: ; Art1 (A/A), Biology (A/A), English 9 (A/A), Precalculus Honors (A/A), Spanish 3 (A/A), Video Production (A/A), World History (A/A)

University of North Dakota, Online Since Fall 2026
[Math 207: Introduction to Linear Algebra](#) (2 credits, in progress)

Stanford University, Stanford, CA Summer 2026
CS106EA: Exploring Artificial Intelligence (3 credits, Grade B+)

[UC Scout](#), () Online Course Fall 2023
JAVA140: Introduction to Java, (5 credits,

SKILLS

Coding: Proficient in Java, Python

Electronics: Advanced in Model making, circuit design, soldering

Robotics: Advanced in Circuitry , PCB design, CAD, Manufacturing and testing, Tech report writing, Software development

SUMMER ACTIVITIES

[Karr Company](#) , Remote Summer 2025

Software Intern,
Worked closely with CTO and wrote code to build a functional prototype designed for real-world testing and shaping the product's core features and user experience, with the opportunity to contribute to both front-end and back-end development.

[Inspirit AI](#), Online Course Summer 2024
Course: Artificial Intelligence Intensive for High School students

AWARDS & HONORS

\$500 1st Award, Patent and Trademark Office Society, qualified participant, Robotics category, individual, tinyurl.com/AdiM-ISEF International Science and Engineering Fair (ISEF), 2026

Qualified participant, Robotics category, individual, California Science Engineering Fair (CSEF), 2026

Best In Show, Senior Division; Engineering Category, ISEF Finalist; Outstanding Electrotechnology Experiment, IEEE; US Airforce ; Santa Cruz County Science Fair, 2026

Distinguished Conrad Innovator Award, Founder/Captain, team of 2, *Fireguard Technology*, tinyurl.com/AdiMConrad, 2025-2026

2025-
Qualified as Observer, International Science and Engineering Fair (ISEF), 2025

Qualified participant, Computer Science category, individual, California Science Engineering Fair (CSEF), 2025

1st place, Computer Science category, individual, Outstanding Electrotechnology, IEEE: \$100 award Keet Family Award; Exceptional Genius Olympiad Award <https://tinyurl.com/AdiM-2025>, Santa Cruz County Science Fair, 2025

INDEPENDENT ENGINEERING ACTIVITIES

10 hours per week, 17 weeks per year

- read, watch, talk...
- concept creator, self-developed engineer,...
- toolsProtoyped hardware, mater

Science Fair Projects

Since 2018

2026 *Design and Development of an Autonomous Wildfire Suppression Firefighting Robot ()*

- Prototyped an autonomous firesensing and extinguishing robot
- Coded a sensor fusion algorithm with embedded system using YOLOv8
- Zoom interview, operational and current issues, Battalion Chief, Robert Culbertson, San Jose, CA

Fireguard Technology: Innovation Stage, captain/founder, team of 2

2025

Title - Using Deep Learning AI to Enable the Safety of Self-Driving Cars

- Developed a purely software based project to compare two CNN models
- Used CNNs to classify road signs for safer autonomous driving.
- The dataset included 3,530 training and 638 testing images across 14 categories.
- Compared VGG-16's architecture with ResNet-50's residual network.
- ResNet-50 achieved 77% accuracy, while VGG-16 achieved only 23%.
- VGG-16 likely overfit because its 138 million parameters exceeded dataset needs.
- Both models maintained similar testing performance, indicating good generalization.

2024

Qualified Participant in Thermo Fisher Scientific Junior Innovator Challenge, 2024

Qualified participant Qualified participant, Physics category, individual, California Science Engineering Fair (CSEF), 2024

1st place, Physics category; Honorable Mention for Best in Show, US Air Force; Most Outstanding Exhibit in Science, Technology Engineering and Mathematics, Yale Science and Engineering Association; Naval Science Award, Office of Naval Research; Best in Innovation and Design, Tech Interactive; Outstanding Electrotechnology Experiment, IEEE; Science Achievement Award, California Association of Professional Scientists
Award: tinyurl.com/AdiM2024; Santa Cruz County Science Fair, 2024

2023

3rd place - Physics category, individual, California Science Engineering Fair (CSEF), 2023

Qualified Participant in Thermo Fisher Scientific Junior Innovator Challenge, 2023

1st place, Physics Category; Outstanding Electrotechnology Experiment, IEEE; Science Achievement Award, California Association of Professional Scientists
Award: tinyurl.com/AdiM-2023, Santa Cruz County Science Fair, 2023

2022

Qualified participant Qualified participant, Physics category, individual, California Science Engineering Fair (CSEF), 2022

Qualified Participant in Thermo Fisher Scientific Junior Innovator Challenge, 2022

1st place, Physics category; Outstanding Electrotechnology Experiment, IEEE; US Metric Association Award
tinyurl.com/AdiM2022; Santa Cruz County Science Fair, 2022

2024 Towards Sensor Fusion

- Developed a multimodal sensor fusion project to precisely locate objects in the field of view
- LiDAR measured reflected laser pulses to create precise, detailed maps of surrounding objects.
- Ultrasonic sensors measure reflected sound waves to detect nearby objects.
- Infrared sensors detect objects using reflected infrared radiation and function well in sunlight.
- Combined digital readouts from the three sensors to create a confirmation of an obstacle on the road.
- Created a test bed to calibrate XY location of the obstacle and triangulate using sensor data

2023 Ultrasonic Eyes

- Self-driving cars developed to test HC-SR04 ultrasonic sensor's accuracy.
- Temperature, materials, and environmental disruptions were evaluated for their effects on readings.

2022 Flying High with Drones

AWARDS

2026 MATE ROV World Championship - 1st place for Hephaestus Robotics Team
-(Category: Ranger, (Won among 47 teams from 15 countries), St Johns, Newfoundland & Labrador, Canada

2026 MATE Northern California Regional ROV Competition - 1st place for Hephaestus Robotics Team -Category: Ranger, Watsonville, CA

2025 MATE ROV World Championship - 3rd place for Hephaestus Robotics Team
-Category: Ranger, Alpena, Michigan, USA

2025 MATE Northern California Regional ROV Competition - 1st place for Hephaestus Robotics Team -Category: Ranger, Watsonville, CA

2024 MATE Northern California Regional ROV Competition - 2nd place for Angry Otters Team -Category: Navigator, Watsonville, CA

TEAM ENGINEERING ACTIVITIES

5 hours per week for 6 months and , ~10 hours for 2 months

X Academy, Hephaestus Robotics Team Santa Cruz, CA

Role- VP Engineering, Team Category - Ranger

- Lead a group of 3 coders , support the CTO in making ROV, built an AI detection system

- In charge of management and production of the main ROV

- Manufacturing and testing, Tech report writing, Software development

2025 - X Academy, Hephaestus Robotics Team Santa Cruz, CA

Role- Electrical Engineer/Circuit Designer , Team Category - Ranger

Custom designed the PCB's for ROV's internal electronics

Technical report writing and Circuitry

Develop an AI recognition software that identifies invasive European green crabs in real time

2024 - X Academy, Angry Ottors Robotics Team Santa Cruz, CA

Team Category - Navigator

- Constructed the inner electronics of the ROV, developed custom PCB boards to maximize ROV productivity, 2024

2023 - X Academy, Angry Ottors Robotics Team Santa Cruz, CA

Team Category - Navigator

- Worked on hardware and ROV design, development, and testing, 2023

2023- The Tech Challenge- Represented Pacific Collegiate School at The Tech Interactive

Hobbies:

Guitar - Self educated guitarist, learn new techniques (finger picking, hammer-offs, etc.) from YouTube, practice tablature of rock songs, self-entertainment, since 2016

Wood working- Amateur carpenter/artist, Carver, Knife Whittling, Sanding and figurines, Beginner at using Dremel

Tennis- Recreational player since 2019

Cooking- Amature cook and recipe creator

Reading -avid reader of fantasy fiction and graphic novels

COMMUNITY ACTIVITIES

Aegis Living - Santa Cruz, CA

May 2026- Present- 30 hours

Help elderly in an assisted living facility with tech needs and recreation hour

Teen Kitchen Project -Santa Cruz, CA

April 2023 - December 2023- 60 hours

Prepared and cooked nourishing meals for Santa Cruz County families facing cancer and other life-threatening illnesses

YouthServe - Santa Cruz, CA

June 2024 - August 2024 - 60 Hours

Volunteered for various community volunteer needs like environmental clean-ups, graffiti removal, native plant restoration, beach clean-up, Food bank food sort and packaging etc.

Pacific Collegiate School - Santa Cruz, CA

June 2023 - August 2026 - 80 Hours

Volunteered for various community and school events needs like State Park clean up, Iron Man event volunteering, School orientation event, Beach clean up, Tutoring for math, Stuff the Bus, Video recording for school events etc.